



# Does DFER-GCViT Hold Up Under Real-World Occlusions?

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## 1. INTRODUCTION

Driver facial expression recognition has never been verified under real-world occlusions.

**Gap:** DFER-GCViT (Saadi et al., 2023) achieves 98.27% on KMU-FED but is never tested under occlusion.

**Why it matters:** Grad-CAM confirms heavy reliance on the mouth region, vulnerable to masks and sunglasses.

**This study:** Synthetic occlusions applied to KMU-FED. Robustness measured via accuracy drop (%).



Fig. 1. Clean vs. synthetically occluded driver face showing the two masking conditions applied to KMU-FED images in this study (Google Gemini).

## 2. RESEARCH QUESTION

Grad-CAM reveals mouth-region dependency, yet occlusion is never tested. That gap drives this study.

**RQ1:** How much does DFER-GCViT's accuracy and F1-score drop under synthetic occlusion at test time?

**RQ2:** Can training with occluded samples reduce that performance gap?

## 3. METHODOLOGY

**Step 1: Baseline Reproduction** Pretrained GC-ViT with modified classifier head (Linear, BatchNorm, ReLU, Dropout, Linear).

**Step 2: Synthetic Occlusion Generation** Proportional bounding boxes applied programmatically. No manual labelling.

**Step 3: Model Training** Model A: clean only. Model B: clean + occluded. Identical held-out test folds.

**Step 4: Robustness Evaluation** Accuracy drop per expression class. Links to Grad-CAM findings.

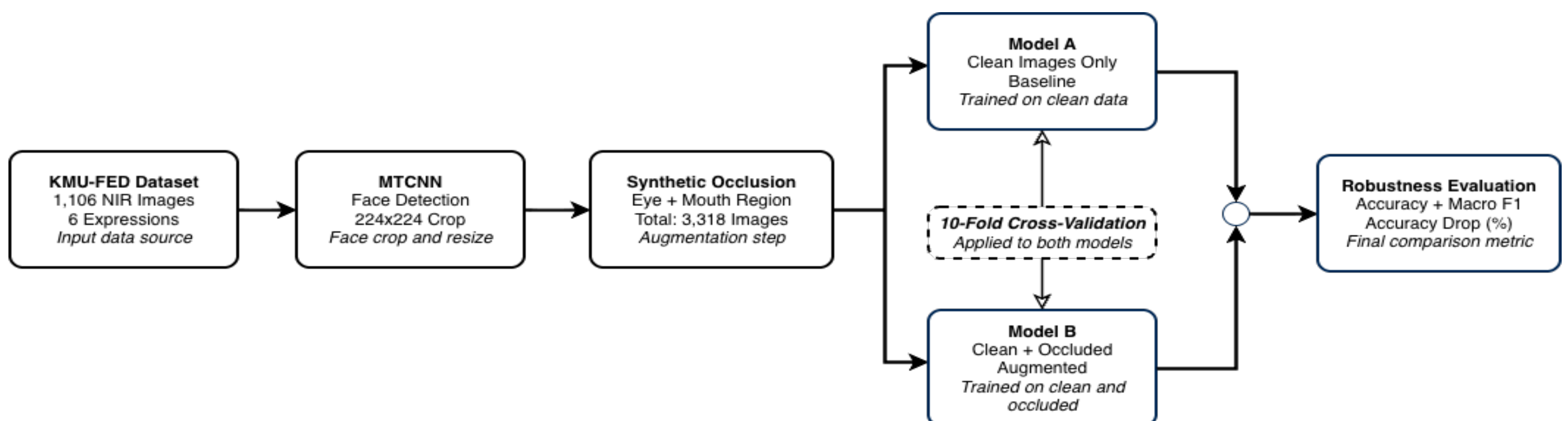


Fig. 2. Proposed pipeline comparing Model A trained on clean images only against Model B trained with occlusion augmentation, evaluated via accuracy drop on KMU-FED (Draw.io).